AISC5000 Foundation of Machine Learning and AI

Homework 4

Due date: 11:59 pm, Fri, Nov 13, 2026

1. Consider the linear model $\widehat{y} = x^{\mathsf{T}} w$, where $x, w \in \mathbb{R}^2$, with two training samples

$$x_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad y_1 = 1, \qquad x_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad y_2 = 3.$$

Define the individual losses and the average training loss by

$$J_n(w) = \frac{1}{2}(x_n^{\mathsf{T}} w - y_n)^2, \qquad J(w) = \frac{1}{2}\big(J_1(w) + J_2(w)\big).$$

   (a) Compute the gradient $\nabla J(w)$ and the stationary point w^* of $J(w)$.

   (b) Compute the Hessian matrix $H = \nabla^2 J(w)$ and its eigenvalues. Use them to classify w^* as a local minimum, local maximum, or saddle point, or state if the second-derivative test is inconclusive.

2. Use the same model and losses as in Question 1. Take $\eta = \frac{1}{2}$ and start each method separately from $w^{(0)} = [0, 0]^{\mathsf{T}}$. For every update below, compute the full-data loss $J(w^{(1)})$.

   (a) Perform one full-batch gradient-descent step

$$w^{(1)} = w^{(0)} - \eta \nabla J(w^{(0)}).$$

   (b) Perform one SGD step using the first training sample, and repeat using the second training sample. In each case, start separately from $w^{(0)} = [0, 0]^{\mathsf{T}}$ and use

$$w^{(1)} = w^{(0)} - \eta \nabla J_n(w^{(0)}), \qquad n = 1, 2.$$

   Compute $w^{(1)}$ and $J(w^{(1)})$ for both cases.

   (c) Perform one full-gradient momentum step with $\mu = \frac{1}{2}$ and $v^{(0)} = [0, 0]^{\mathsf{T}}$:

$$v^{(1)} = \mu v^{(0)} - \eta \nabla J(w^{(0)}), \qquad w^{(1)} = w^{(0)} + v^{(1)}.$$

   Compute $v^{(1)}$, $w^{(1)}$, and $J(w^{(1)})$.

3. **Programming: momentum.** Write a Python program that applies full-gradient momentum to $J(w)$ from Question 1. Use $w^{(0)} = v^{(0)} = [0, 0]^{\mathsf{T}}$, $\eta = \frac{1}{2}$, and $\mu = \frac{1}{2}$, as in Question 2(c), and perform 20 updates:

$$v^{(t+1)} = \mu v^{(t)} - \eta \nabla J(w^{(t)}), \qquad w^{(t+1)} = w^{(t)} + v^{(t+1)}.$$

Print the iteration number, both coordinates of $w^{(t)}$, and $J(w^{(t)})$ for $t = 0, 1, 2, 5, 10, 20$. (NumPy or Pytorch may be used. You can use one the following way for the submission of the code and output presentation: (i) Latex; (ii) Screenshot; (iii) Copy code and outputs to your submitted file; (iv) Hand-writing.)